Thus the estimate for the change in the volume is 40π cm^3.

Exact calculation of the volume change gives

$$V(10.1) - V(10) = 2040.2\pi - 2000\pi = 40.2\pi \text{ cm}^3.$$

So relative error $= \dfrac{|40.2\pi - 40\pi|}{40.2\pi} = \dfrac{1}{201} = 0.00497$; and hence the percentage error = relative error $\times 100 = \dfrac{1}{201} \times 100 = 0.497\%$.

EXERCISE 8.1

1. Let $f(x) = \sqrt[3]{x}$. Find the linear approximation at $x = 27$. Use the linear approximation to approximate $\sqrt[3]{27.2}$.
2. Use the linear approximation to find approximate values of
   (i) $(123)^{\frac{2}{3}}$ (ii) $\sqrt[4]{15}$ (iii) $\sqrt[3]{26}$
3. Find a linear approximation for the following functions at the indicated points.
   (i) $f(x) = x^3 - 5x + 12,\ x_0 = 2$ (ii) $g(x) = \sqrt{x^2 + 9},\ x_0 = -4$
   (iii) $h(x) = \dfrac{x}{x+1},\ x_0 = 1$
4. The radius of a circular plate is measured as 12.65 cm instead of the actual length 12.5 cm. find the following in calculating change in the area of the circular plate:
   (i) Absolute error (ii) Relative error (iii) Percentage error
5. A sphere is made of ice having radius 10 cm. Its radius decreases from 10 cm to 9.8 cm. Find approximations for the following:
   (i) change in the volume (ii) change in the surface area
6. The time T, taken for a complete oscillation of a single pendulum with length l, is given by the equation $T = 2\pi\sqrt{\dfrac{l}{g}}$, where g is a constant. Find the approximate percentage error in the calculated value of T corresponding to an error of 2 percent in the value of l.
7. Show that the percentage error in the nth root of a number is approximately $\dfrac{1}{n}$ times the percentage error in the number

8.2.3 Differentials

Here again, we use the derivative concept to introduce **"Differential"**. Let us take another look at (1),

$$\frac{df}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = f'(x) = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x}. \qquad \ldots(7)$$

Here $\dfrac{df}{dx}$ is a notation, used by Leibniz, for the limit of the difference quotient, which is called the **differential coefficient** of y with respect to x. Will it be meaningful to treat $\dfrac{df}{dx}$ as a quotient of df and dx? In other words, is it possible to assign meaning to df and dx so that derivative is equal to

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